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P8465

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Oct-22/BE/Insem-41

B.E. (Computer Engineering)

DESIGN AND ANALYSIS OF ALGORITHMS

(2019 Pattern) (Semester - VII) (410241)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer the question of Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Why correctness of the algorithm is important? Define loop invariant property and prove the correctness of finding summation of n numbers using loop invariant property. **[8]**

b) What is iterative algorithm? Explain iteractive algorithm design issues using examples. **[7]**

OR

Q2) a) How to prove that an algorithm is correct? How to prove the correctness of an algorithm using counter example? Give suitable example. **[7]**

b) Write a short note on any 4 problem solving strategies. **[8]**

Q3) a) What is Best, Average and Worst case Analysis of Algorithms? Analyse the following algorithm Best, Average and Worst case **[8]**

```
void sort (int a. int n) {
   int i, j;
   for (i = 0; i < n; i++) {
      j = i-1;
      key = a[i];
      while (j >=0 && a[j] > key)
      {
         a[j+1] = a[j];
         j = j-1;
      }
      a[j+1] = key;
   }
}
```

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b) • Explain P, NP, NP-Hard and NP-Complete problems with examples.

• Explain 3-SAT problem using an example. Why is SAT so important in theoretical computer science?

[7]

OR

Q4) a) What is NP-complete class problem? How would you prove vertex cover problem is NP-complete class problem? **[8]**

b) What is Best, Average and Worst case Analysis of Algorithms? Analyse the following algorithm Best, Average and Worst case **[7]**

```
int Linear-search(int a, int n, int item) {
  int i;
  for (i = 0; i < n; i++) {
  if (a[i] = = item) {
        return a[i]
    }
  }
  return - 1
```

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Study material provided by: Vishwajeet Londhe

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[6188]-141

B.E. (Computer Engineering) (Insem)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester-VII) (410241)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates:

1) ***Answer the questions Q.1 or Q.2; Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indictae full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Given the fastest computer and hypothetically infinite memory, do we still need to study algorithms? Justify. **[2]**

b) How can we related algorithms to technology? Briefly explain. **[6]**

c) Consider an array A of n integers which are already in sorted order. Let x be the number being searched in the array A in a liner fashion. The code fragment performing this task is given below: **[7]**

```
int lin _ search (int A [])
{
    i=0; flag=0;
    do { if (x = = A [i]) then
        return (1); // Number found
        else
            i++;
} while (i<n);
    return (0); // Number not found.
}
```

i) Is this code fragment efficient? (We wish to use linear search only). Justify your answer.

ii) Does it attribute to any design issue with respect to iterative algorithm? Briefly explain.

OR

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Q2) a) What is iterative algorithm? Explain interative algorithm design issues using suitable examples. **[8]**

b) Consider the following algorithm to find the square of a number:

```
int sqr(int n)
    {
        if (n= = 0) return 0;
        else return (2n+sqr(n-1)-1)
```

Prove the correctness of this algorithm using principle of mathematical induction or otherwise. **[7]**

Q3) a) Briefly explain P and NP problems in the context of complexity theory. Give suitable example. **[8]**

b) If f (n)=O(g(n)) then does it imply g(n)=O(f(n))? Discuss. **[5]**

c) Comment on the statement "Best case analysis of algorithm may not give clear idea of performance" **[2]**

OR

Q4) a) What is SAT AND 3-SAT problem? Prove that 3-SAT problem is NP complete. **[8]**

b) What to do you understand by best case, worst case and average-case behaviour of an algorithm? Is an average case efficiency an average of best-case, worst-case efficiencies? Justify answer. **[7]**

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B.E. (Computer Engineering) (Insem)

DESIGN AND ANALYSIS OF ALGORITHMS

(2019 Pattern) (Semester - VII) (410241)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figure to right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Prove the correctness of the following algorithm, which finds the minimum number from the given array of n numbers.

```
int min_element (int A [ ])
{
        min=A[0];
        for (i=1; i<n ; i++)
        {
          if (A[i]<min) then
              min=A[i];
        }
        return(min);
}
```

[7]

b) What is iterative algorithm? Explain iterative algorithm design issues using examples. **[8]**

OR

Q2) a) Prove the correctness of the following algorithm, which finds the factorial of a given number.

```
int fact (int n)
{ if (n==0) then return 1;
  else
      return (n * fact(n-1));
}
```

[7]

b) Write a short note on any four problem solving strategies. **[8]**

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Q3) a) Consider the algorithm for selection sort:

```
void  select (int A[ ])
 {// A is an array of n numbers.
  for (i=1;i<n-1; i++)
  {
      min_pos=i;
      min_val=A[i];
      for(j=i+1; j<=n;j++)
      {
         if(A[j] <min_val) then
         {
           min_pos=j;
            min_val=A[j];
          }
       }
      A[min_post]=A[i];
      A[i]=min_val;
   }
  }
```

Analyse the complexity of this algorithm. Clearly indicate the assumptions made, if any. **[6]**

b) If an algorithm has a time complexity of both O(f(n)) and Ω (g(n)) for the same function f(n) and g(n), then what does it imply? Explain. **[3]**

c) What do Ω and Θ notations mean? When do we use O notation? **[6]**

OR

Q4) a) What is polynomial time reducibility? What is its importance in computational complexity theory? **[6]**

b) What do understand by NP complete and NP hard problems? Give examples. **[6]**

c) Is $6n^3 = \Theta(n^2)$? Justify your answer. **[3]**

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